

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard

Question

Rectangles $ABCD$ and $EFGH$ are similar. The length of each side of $EFGH$ is **6** times the length of the corresponding side of $ABCD$. The area of $ABCD$ is **54** square units. What is the area, in square units, of $EFGH$?

Answer

- A. **9**
- B. **36**
- C. **324**
- D. **1,944**

Correct Answer: D

Rationale

Choice D is correct. The area of a rectangle is given by bh , where b is the length of the base of the rectangle and h is its height. Let x represent the length, in units, of the base of rectangle $ABCD$, and let y represent its height, in units. Substituting x for b and y for h in the formula bh yields xy . Therefore, the area, in square units, of $ABCD$ can be represented by the expression xy . It's given that the length of each side of $EFGH$ is **6** times the length of the corresponding side of $ABCD$. Therefore, the length, in units, of the base of $EFGH$ can be represented by the expression $6x$, and its height, in units, can be represented by the expression $6y$. Substituting $6x$ for b and $6y$ for h in the formula bh yields $(6x)(6y)$, which is equivalent to $36xy$. Therefore, the area, in square units, of $EFGH$ can be represented by the expression $36xy$. It's given that the area of $ABCD$ is **54** square units. Since xy represents the area, in square units, of $ABCD$, substituting **54** for xy in the expression $36xy$ yields $36(54)$, or **1,944**. Therefore, the area, in square units, of $EFGH$ is **1,944**.

Choice A is incorrect. This is the area of a rectangle where the length of each side of the rectangle is $\sqrt{\frac{1}{6}}$, not **6**, times the length of the corresponding side of $ABCD$.

Choice B is incorrect. This is the area of a rectangle where the length of each side of the rectangle is $\sqrt{\frac{2}{3}}$, not **6**, times the length of the corresponding side of $ABCD$.

Choice C is incorrect. This is the area of a rectangle where the length of each side of the rectangle is $\sqrt{6}$, not **6**, times the length of the corresponding side of $ABCD$.